Leveraging G×E interaction to optimize multi-trait selection in Lima Bean

Gérson N.C. Ferreira[1]†* João P.S. Pavan[1]† Mauricio S. Araújo[1]
Dayana R. Sousa[1] Michelle S. Nascimento[2] Vanessa G. Moura[2]
José T. B. Chagas[1] Yasmim I. Retore[1] Josieli L. Silva[2]
Maria S. S. Silva[2] Regina L.F. Gomes[2] Ângela C.A. Lopes[2]
Maria I. Zucchi[3,4] José B. Pinheiro[1]*

2026-05-05

Affiliations:

[1] Genetics Diversity and Breeding Laboratory, Department of Genetics, "Luiz de Queiroz" College of Agriculture (ESALQ/USP), Piracicaba, São Paulo, Brazil

[2] Department of Plant Science, Federal University of Piauí, Teresina, Piauí, Brazil

[3] Conservation Genetics and Genomics Group, Department of Genetics, "Luiz de Queiroz" College of Agriculture (ESALQ), University of São Paulo (USP), Piracicaba, São Paulo, Brazil

[4] Secretariat of Agriculture and Food Supply of São Paulo State, Piracicaba, São Paulo, Brazil

Corresponding authors * gerson.nascimento@live.com (G.N.C. Ferreira) and jbaldin@usp.br (J.B. Pinheiro)

† *These authors contributed equally to this work*

```r
load("../Data/ebvMat.rda")
```

1 Desire index

```r
# DESIRE INDEX FUNCTION

gainDiff <- function(ebvMat, b_ebv, dGain, nSel){

  take <- order(ebvMat %*% b_ebv, decreasing = TRUE)[1:nSel]

  mu_Sel <- colMeans(ebvMat[take, ])
  resp   <- mu_Sel - colMeans(ebvMat)
  resp   <- resp / sqrt(sum(resp^2))

  sum((resp - dGain)^2)
}
```

```r
ebvMat <- ebvMat[complete.cases(ebvMat), ]
#COV MATRIX -EBVS
library(AlphaSimR)
```

```
## Loading required package: R6
```

```r
G_ebv <- AlphaSimR::popVar(ebvMat)
G_ebv
```

```
##            [,1]       [,2]       [,3]       [,4]       [,5]       [,6]
## [1,] 6587.97833 217.258861 424.765720 256.943342 36.6766871 4080.22632
## [2,]  217.25886   7.323476  14.181373   8.521450  1.2173849  134.59974
## [3,]  424.76572  14.181373  28.804036  17.134163  2.4397036  266.63926
## [4,]  256.94334   8.521450  17.134163  10.367730  1.4703171  160.61068
## [5,]   36.67669   1.217385   2.439704   1.470317  0.2129143   22.92692
```

```
## [6,] 4080.22632 134.599743 266.639262 160.610683 22.9269219 2564.70381
```

```r
#desired gains
desiredGain <- c(
  GY   =  2,
  OHSW =  1,
  NDF  = -1,
  NDM  = -1,
  PH   =  0,
  PN   =  1
)
desiredGain <- desiredGain / sqrt(sum(desiredGain^2))
sum(desiredGain^2)
```

```
## [1] 1
```

```r
b_ebv <- solve(G_ebv) %*% desiredGain
b_ebv
```

```
##             [,1]
## [1,] -0.07014002
## [2,]  2.70032556
## [3,] -0.38420756
## [4,] -2.08437923
## [5,]  8.33887739
## [6,]  0.06593790
```

```r
nSel <- 5
opt <- optim(
  par     = b_ebv,
  fn      = gainDiff,
  ebvMat  = ebvMat,
  dGain   = desiredGain,
```

```r
  nSel    = nSel
)
opt_weights <- opt$par
opt_weights
```

```
##             [,1]
## [1,] -0.06570972
## [2,]  2.70035237
## [3,] -0.38474376
## [4,] -2.08393017
## [5,]  8.33759052
## [6,]  0.06765372
```

```r
DesireIndex <- ebvMat %*% opt_weights

DesireIndex <- data.frame(
  GEN = rownames(ebvMat),
  DesireIndex = as.numeric(DesireIndex)
) |>
  dplyr::arrange(desc(DesireIndex))

head(DesireIndex, 10)
```

```
##    GEN DesireIndex
## 1  L06    234.5246
## 2  L27    234.2351
## 3  L21    233.6062
## 4  L20    233.5220
## 5  L13    233.4020
## 6  L34    233.3338
## 7  L14    232.8109
```

```
## 8  L11    232.7500
## 9  L25    232.4086
## 10 L12    232.2843
```

```r
selected <- DesireIndex[1:5, ]
selected
```

```
##   GEN DesireIndex
## 1 L06    234.5246
## 2 L27    234.2351
## 3 L21    233.6062
## 4 L20    233.5220
## 5 L13    233.4020
```